

Course number	0418-425	Course Title	Distributed and Parallel Systems
Credit hours	3 (3:0:3)	Course Coordinator	Dr. Hosam AboElfotoh
Compulsory	No	Prerequisite	0418-331

Textbook

<i>Title:</i>	<i>Distributed Systems: Principles and Paradigms</i>
<i>Author:</i>	<i>Andrew S. Tanenbaum</i>

Major Topics Covered in the Course

	Topic	Hours
1	Introduction to distributed and parallel systems: definition, goal, types. Comparison between parallel and distributed systems.	4
2	Distributed Systems architecture and Models.	4
3	Models of Parallel Computers and interconnection networks.	4
4	Concurrent processes; Clients/Servers processes; Inter-process Communication.	6
5	Synchronization; Clock synchronization, Logical clocks, Mutual exclusion, Leader election algorithms.	6
6	Consistency and Replication.	6
7	Parallel algorithms design and analysis.	4
8	Example distributed systems and middleware software.	2
9	Example of parallel architectures (GPU and Cuda).	3
10	Midterm I and Midterm II.	3
	Total.	42

Grading

15-20%	Midterm 1
15-20%	Midterm 2
30-20%	Hmk/ quiz/project
40%	Final

Course Outcomes

Students should be able to:

1. Describe the need for, and, concepts of distributed and parallel systems.
2. Demonstrate the knowledge of potential run-time problems arising from the concurrent operation of many separate processes.
3. Demonstrate the ability of analyzing distributed and parallel processing problems and designing distributed or parallel solutions for them.
4. Compare and contrast different approaches to distributed systems process synchronization.
5. Demonstrate the knowledge of methods for preserving consistency of distributed and parallel systems.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1	L					
2	H	M				M
3	H					M
4	M	M				H
5	M	M				H